

xAI-Proj-B

Bachelor Project

Chair of Explainable Machine Learning (xAI)

Sebastian Doerrich, M.Sc., Prof. Dr. Christian Ledig

xAI Lab Bamberg

University of Bamberg

Introduction

Who are we as a chair?

Who am I?

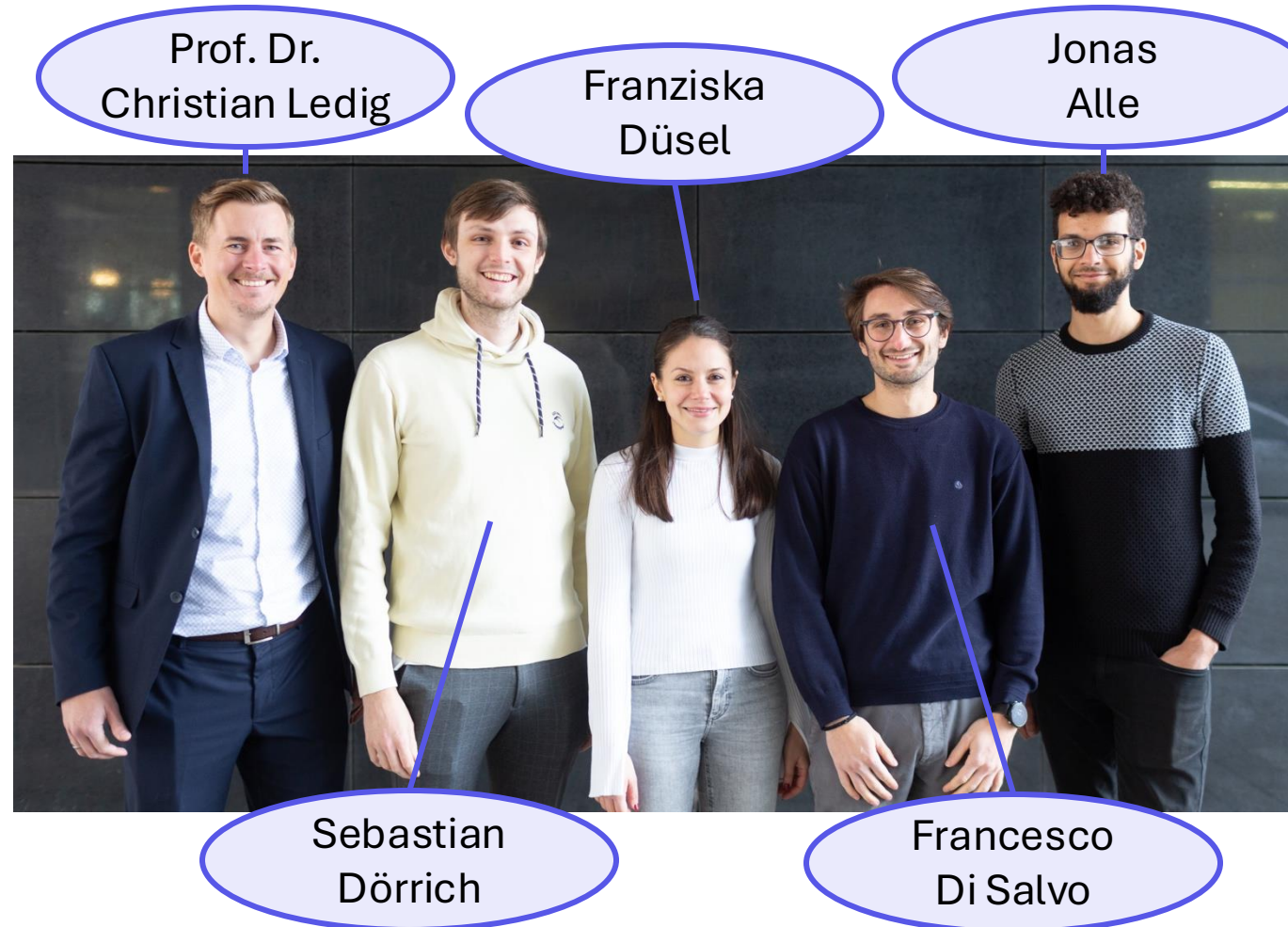
Who are you?



Chair of Explainable Machine Learning (xAI)



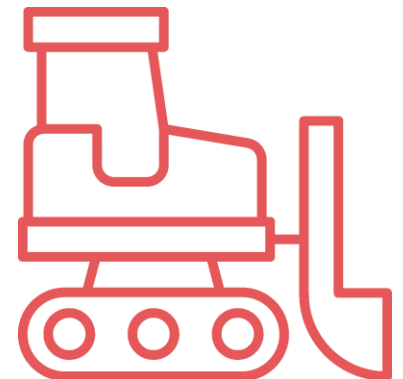
Chair of Explainable Machine Learning (xAI)



Chair of Explainable Machine Learning (xAI) – What do we do?



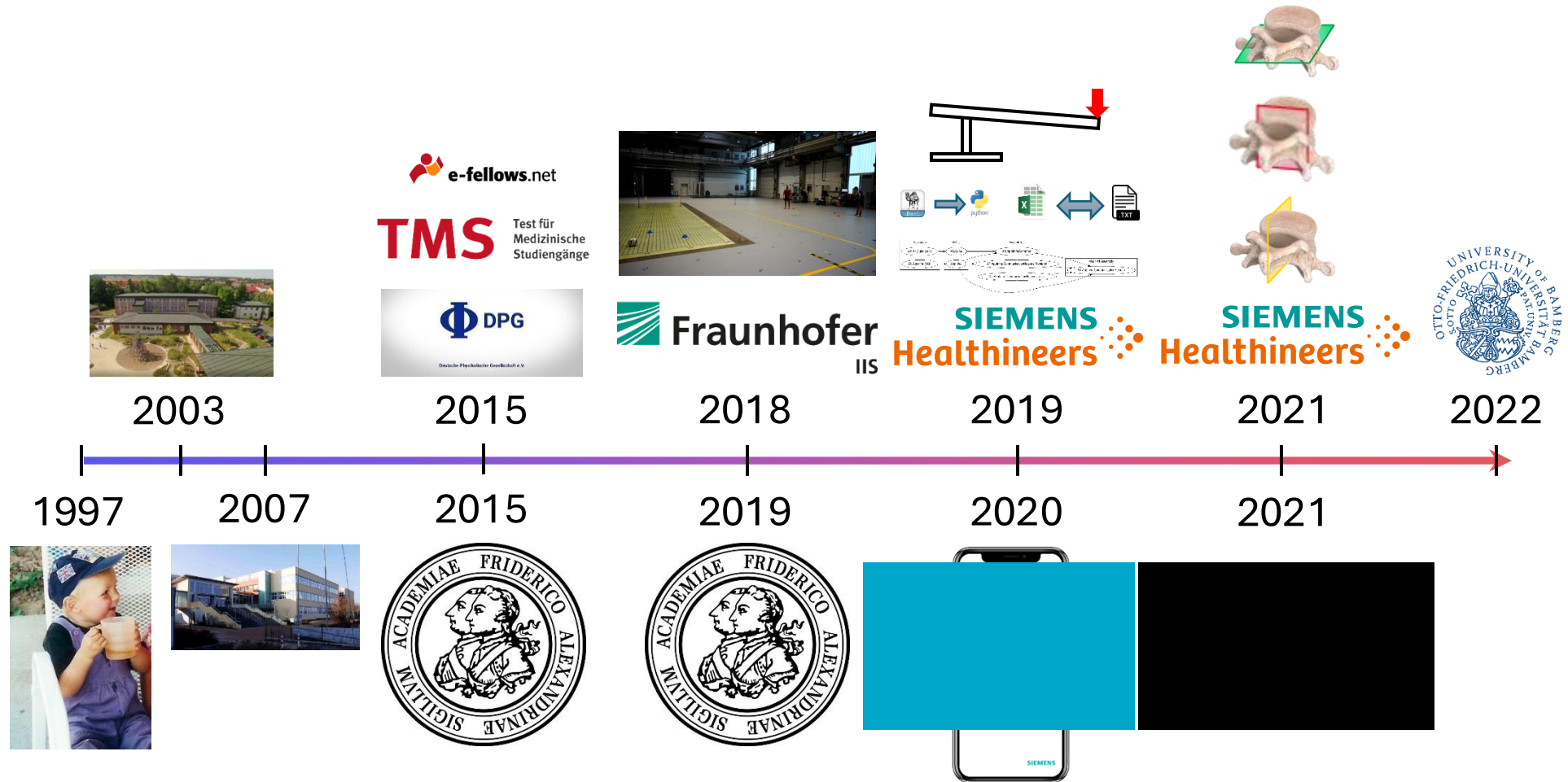
- Development of robust AI algorithms (classification, segmentation, ...)
- Translation of explainable AI into impactful applications
- Data-driven analysis and standardization of large and heterogeneous datasets with focus on data biases
- Development of statistical methods for sampling representative datasets
- Validation of AI systems with focus on generalizability and data biases



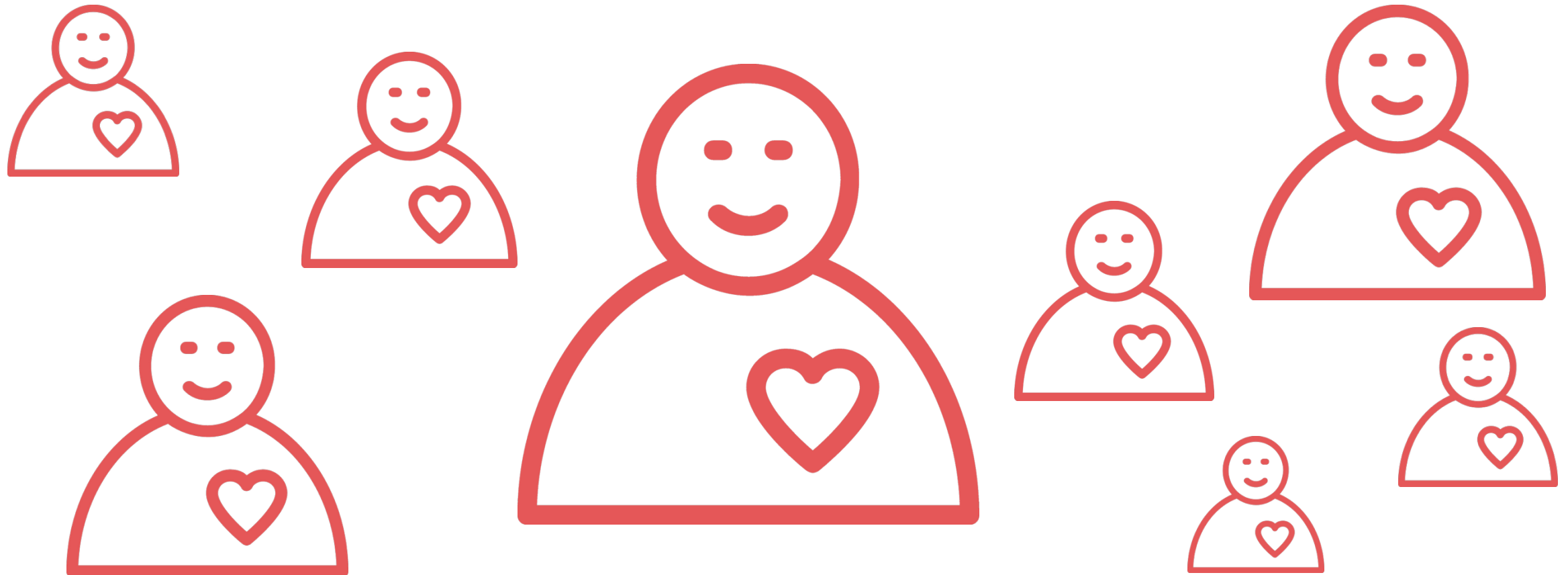
Who am I?



Who am I?



Who are you?



Research

What is my own research about?

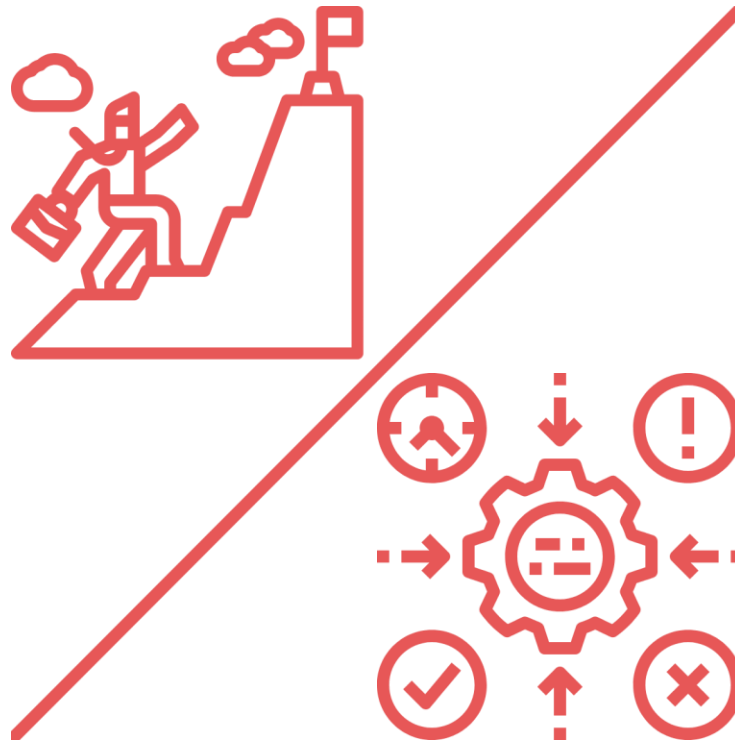
What interests me and my colleagues?

What do we want to achieve?



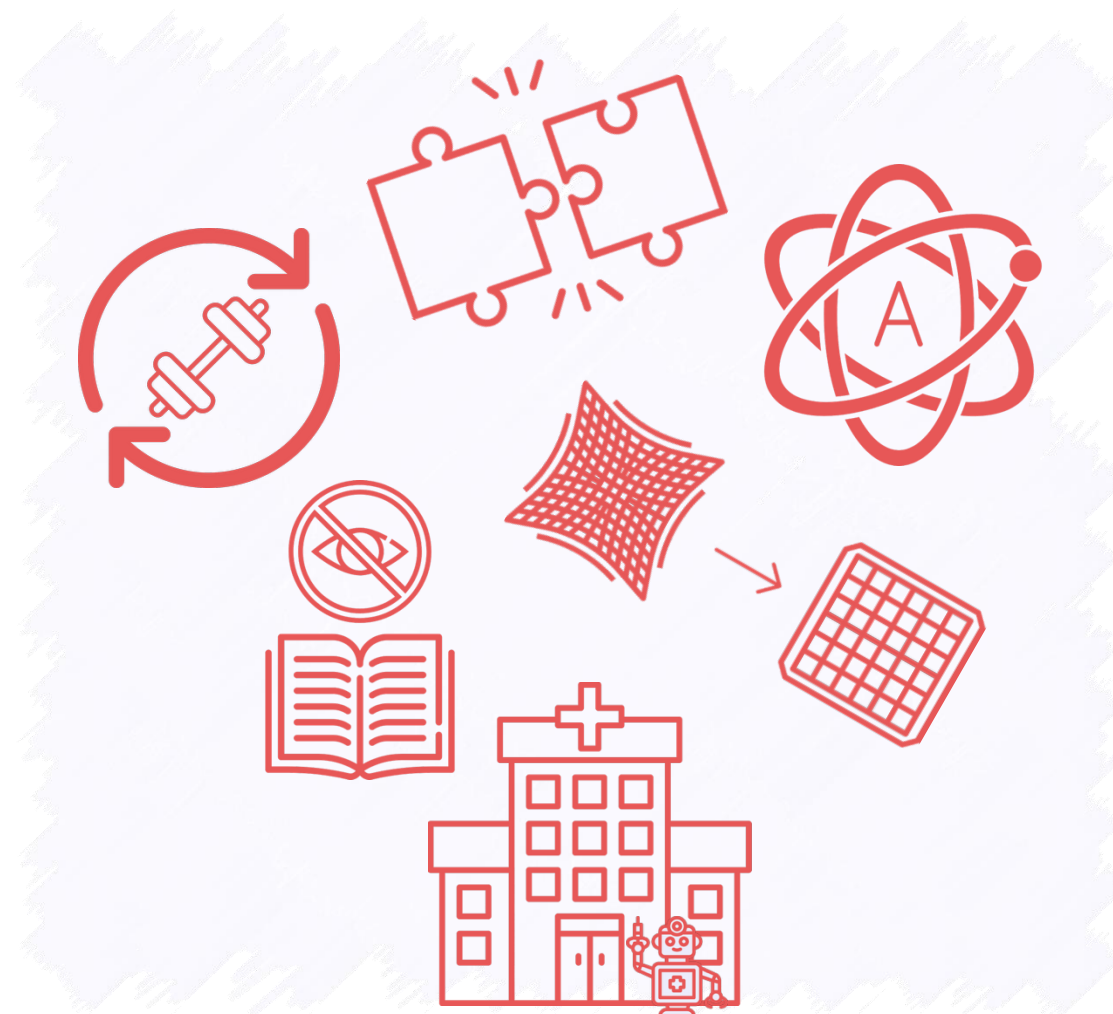
Disentangling Style and Anatomy: A Modality-Agnostic Approach to Robust and Generalizable Medical Image Classification

Motivation & Context

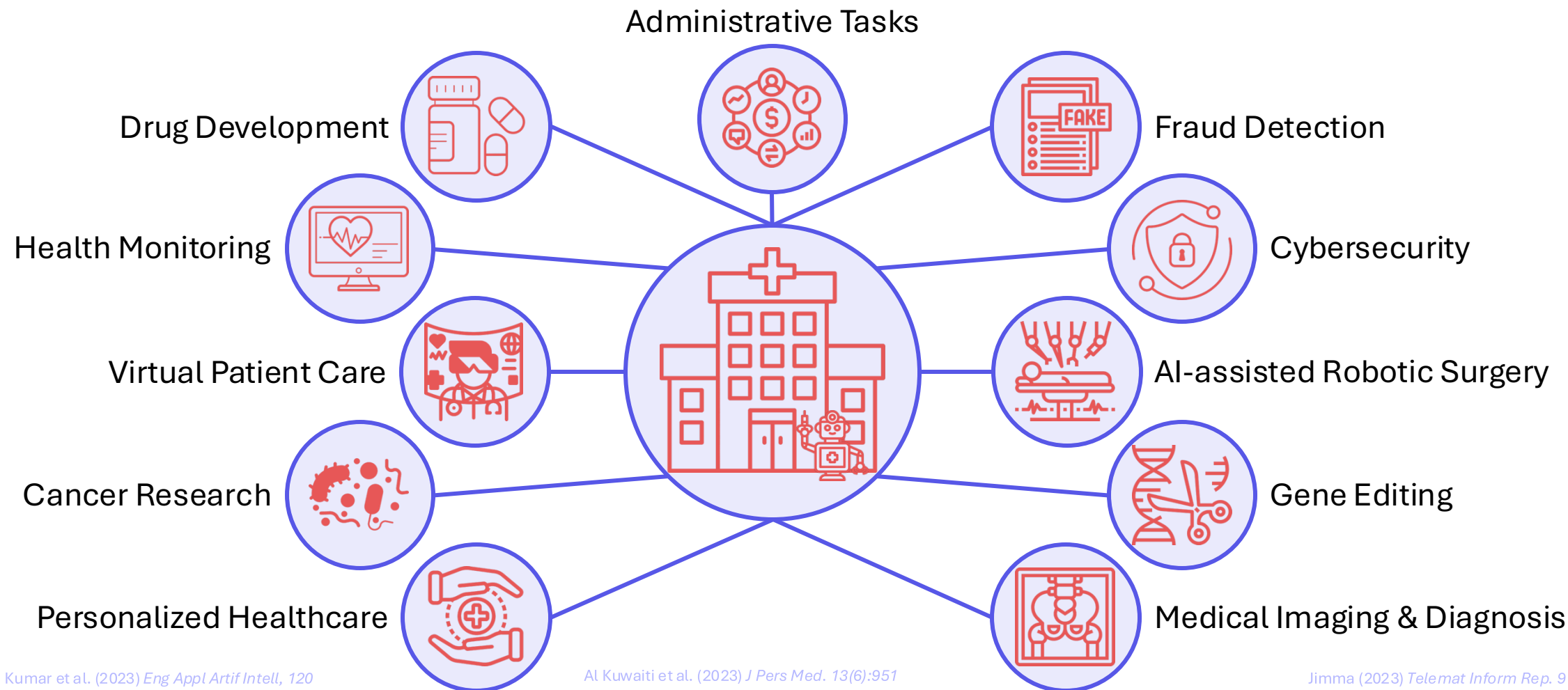


Categorization of Core Concepts

- Feature disentangling
 - Modality-agnostic
 - Robust and generalizable
 - Distortion-invariant
 - Self-supervised
 - Medical image classification
- With Deep Learning!



Areas of Impact for DL in Healthcare



Kumar et al. (2023) *Eng Appl Artif Intell*, 120

Al Kuwaiti et al. (2023) *J Pers Med*. 13(6):951

Jimma (2023) *Telemat Inform Rep*. 9

Types of Medical Data with DL

- Video (e.g., Ultrasound)
- Text (e.g., anamnesis reports)
- Image (e.g., X-ray)
- Numerical (e.g., blood reports)
- Categorical (e.g., test results)
- Speech (e.g., doctor instructions)
- Signals (e.g., ECG)



Inherent Challenges with Medical Data

Ideal



Real

- Large, diverse, high-quality datasets
- Homogeneous data acquisition
- Equal representation of disease classes and severities
- Equal representation of demographics
- Consistent imaging parameters and scanner models.

Zhou et al. (2024) *IEEE Conf Artif Intell Test.* 120-131

- Scarce and distorted data

Upadhyay et al. (2024) *Arch Comput Methods Eng.* 31:1701-1719

- Inhomogeneous data acquisition

Lafarge et al. (2017) *Lecture Notes in CS.* 10553

- Underrepresented disease classes and severities

Afrose et al. (2022) *Commun Med.* 2:111

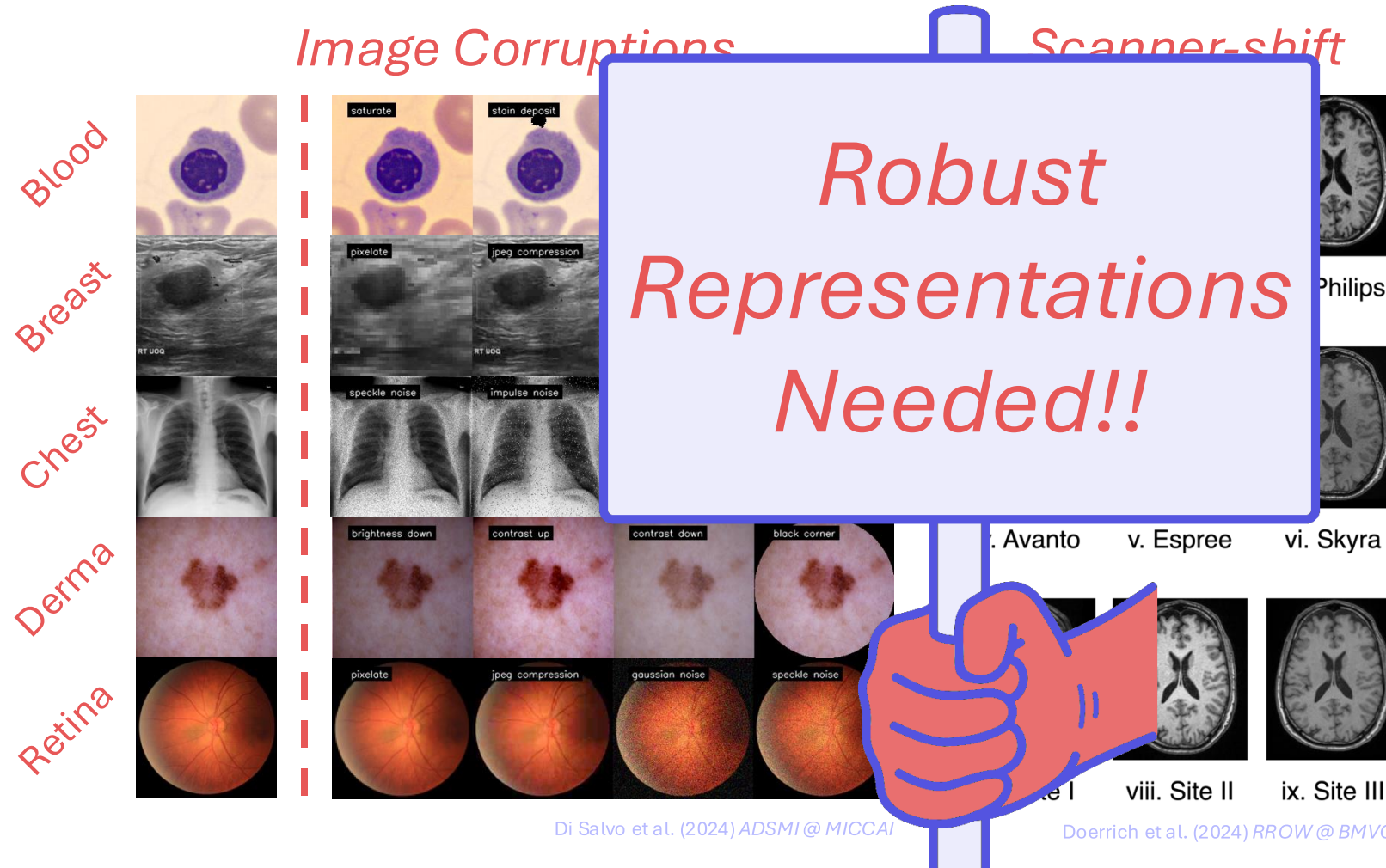
- Underrepresented demographic groups

Moore et al. (2023) *Nature.* 616:259-265

- Scanner-induced domain shifts

Stacke et al. (2021) *IEEE J Biomed Health Inform.* 25:325-336

Distorted / Corrupted Data Domain Shifts



Hospital-shift

	Train			Val	Test
	Hospital 1	Hospital 2	Hospital 3	Hospital 4	Hospital 5
Normal					
Tumor					

Doerrich et al. (2024) MICCAI. 15010

Processing-shift

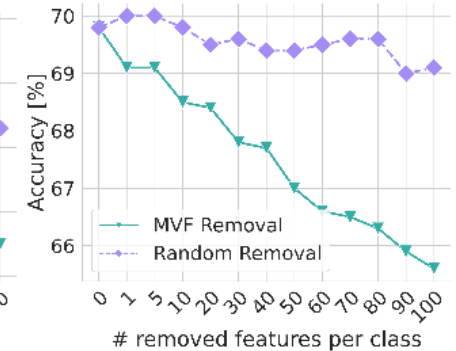
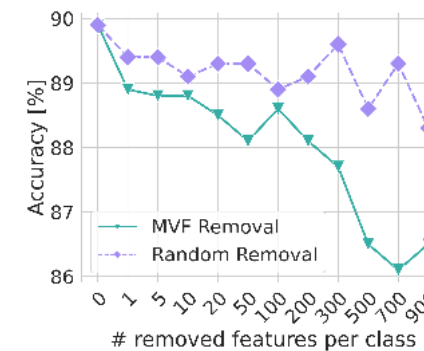
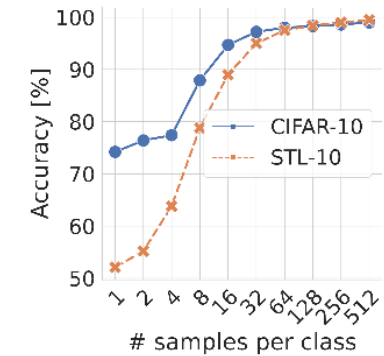
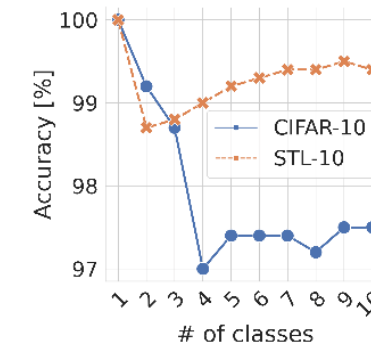
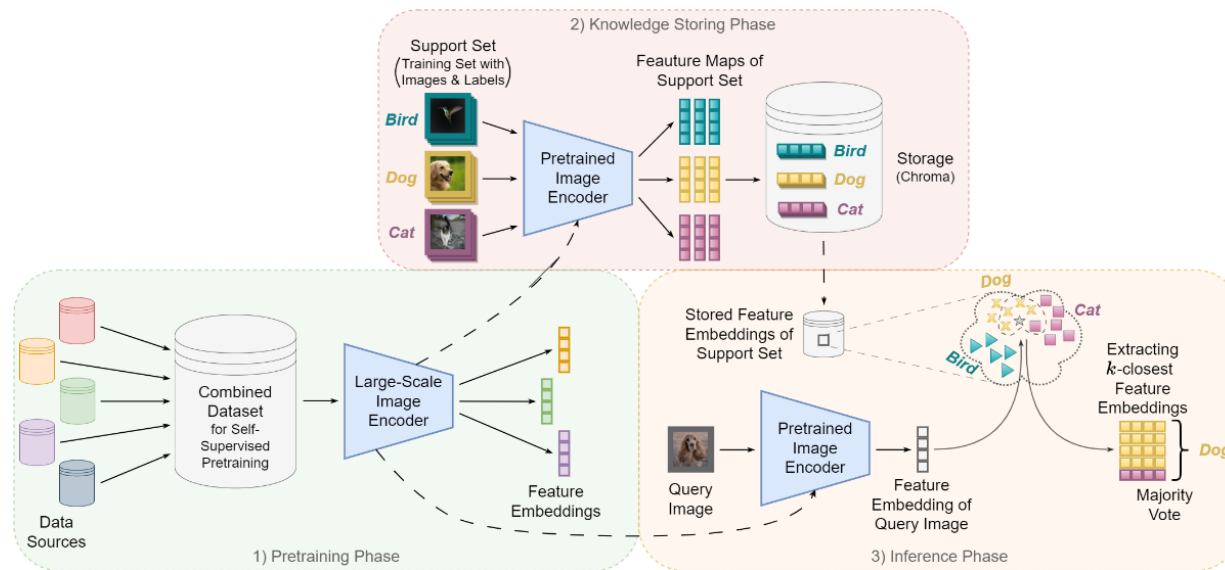
	NKI	VGH	IHC
Epithelium			
Stroma			

Doerrich et al. (2024) MICCAI. 15010

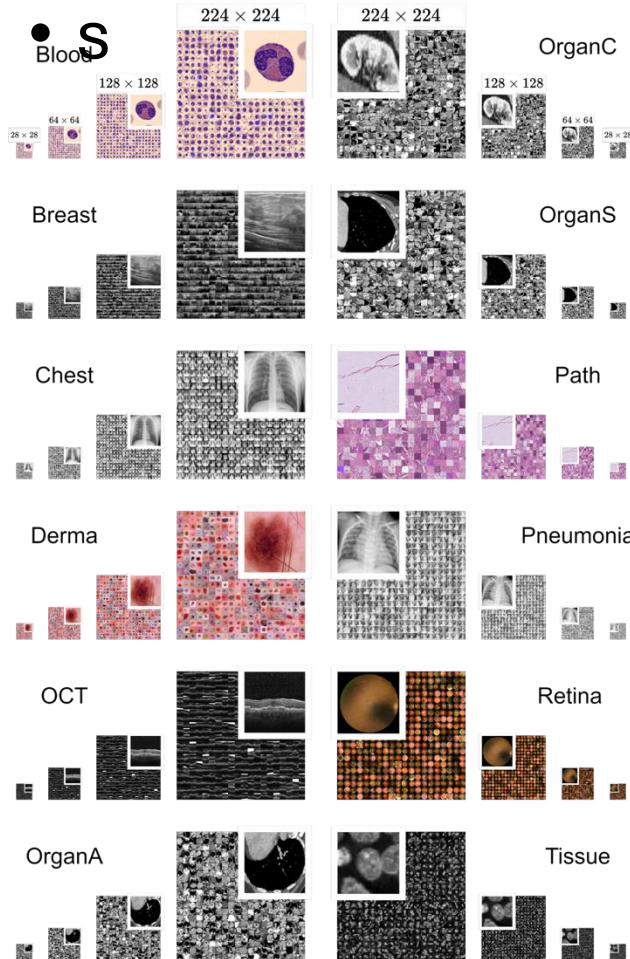
Other Projects – Mine



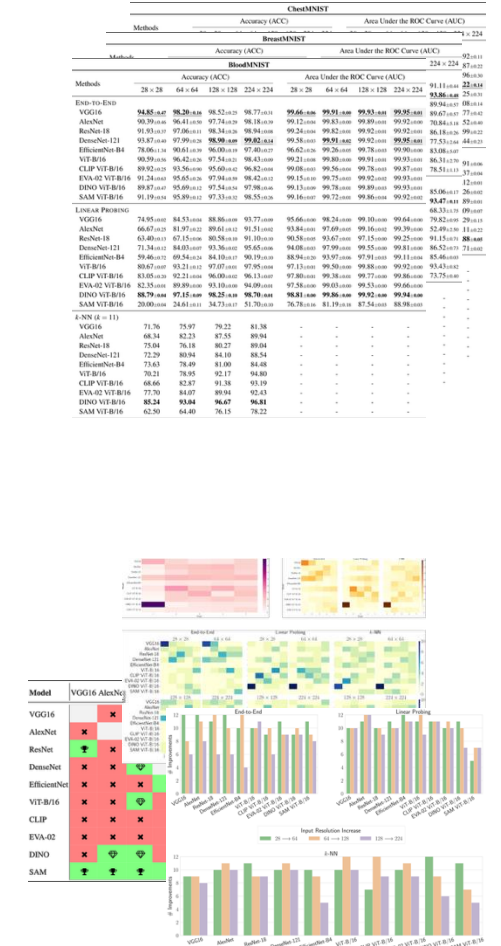
Integrating kNN with Foundation Models for Adaptable and Privacy-Aware Image Classification



Rethinking model prototyping through the MedMNIST+ dataset collection



Methods	Accuracy (ACC)				Area Under the ROC Curve (AUC)			
	28 × 28	64 × 64	128 × 128	224 × 224	28 × 28	64 × 64	128 × 128	224 × 224
END-TO-END								
VGG16	82.34±0.88	85.33±1.16	86.64±0.82	86.70±0.93	92.66±0.41	94.24±0.28	95.16±0.27	95.30±0.22
AlexNet	78.92±0.81	82.94±0.78	85.04±0.74	85.74±0.64	91.14±0.43	92.72±0.33	94.29±0.30	94.90±0.23
ResNet-18	79.66±0.74	83.42±0.65	85.73±0.66	86.22±0.58	90.92±0.28	92.49±0.50	93.91±0.27	94.51±0.24
DenseNet-121	80.32±0.93	84.62±0.80	87.13±0.56	87.11±0.64	91.75±0.55	93.59±0.23	94.57±0.21	95.03±0.23
EfficientNet-B4	73.18±1.61	79.37±1.10	82.52±0.79	82.44±1.11	87.04±0.82	90.07±0.65	91.89±0.39	91.64±0.73
ViT-B/16	78.23±0.88	83.17±0.92	84.94±0.93	86.06±0.92	90.54±0.47	92.53±0.69	93.25±0.35	94.08±0.38
CLIP ViT-B/16	76.73±0.80	80.39±0.99	82.33±1.02	82.75±1.01	89.22±1.11	90.91±0.51	91.51±0.31	91.83±0.58
EVA-02 ViT-B/16	76.69±1.44	80.77±0.97	82.76±0.97	84.72±1.09	88.91±1.01	90.53±0.54	91.59±0.75	92.60±0.94
DINO ViT-B/16	78.51±1.09	82.13±1.02	84.31±1.05	84.84±1.08	91.02±0.48	91.94±0.32	92.91±0.36	93.90±0.73
SAM ViT-B/16	78.26±0.84	82.13±1.12	84.19±1.06	84.30±0.82	89.36±0.79	90.79±1.01	91.94±1.08	91.91±0.55
LINEAR PROBING								
VGG16	71.18±0.13	75.14±0.26	78.58±0.15	79.62±0.18	87.70±0.06	89.55±0.06	91.94±0.04	92.47±0.05
AlexNet	69.63±0.35	76.11±0.25	79.08±0.17	81.02±0.18	85.91±0.13	89.06±0.11	91.78±0.07	93.18±0.05
ResNet-18	64.41±0.05	70.49±0.08	74.94±0.07	76.89±0.08	82.95±0.18	85.24±0.22	87.97±0.76	90.37±0.17
DenseNet-121	72.10±0.28	78.01±0.19	80.77±0.19	82.22±0.23	86.76±0.94	90.11±0.29	92.00±0.19	93.02±0.11
EfficientNet-B4	67.29±0.54	73.95±0.51	76.39±0.25	77.91±0.62	83.67±0.64	86.48±0.45	88.49±0.55	89.57±0.22
ViT-B/16	73.21±0.24	79.62±0.33	83.08±0.70	84.01±0.27	88.25±0.14	91.33±0.22	93.57±0.09	94.31±0.13
CLIP ViT-B/16	74.17±0.24	78.67±0.32	81.54±0.24	82.24±0.18	88.66±0.08	91.48±0.20	93.28±0.10	93.66±0.10
EVA-02 ViT-B/16	73.04±0.19	76.50±0.16	78.48±0.11	79.30±0.10	88.41±0.04	90.66±0.03	92.04±0.04	92.41±0.04
DINO ViT-B/16	78.23±0.22	82.74±0.36	84.46±0.24	85.11±0.55	90.94±0.14	93.29±0.12	94.50±0.11	94.99±0.14
SAM ViT-B/16	43.69±0.01	48.10±0.03	54.14±0.04	61.20±0.03	66.51±1.38	74.68±0.80	80.46±0.13	81.75±0.11
k-NN (k = 11)								
VGG16	66.07	70.65	72.26	73.78	-	-	-	-
AlexNet	67.47	72.14	74.56	76.22	-	-	-	-
ResNet-18	66.98	71.42	74.20	76.60	-	-	-	-
DenseNet-121	67.12	71.29	74.97	76.97	-	-	-	-
EfficientNet-B4	68.15	72.45	73.91	74.17	-	-	-	-
ViT-B/16	65.92	70.90	75.45	77.51	-	-	-	-
CLIP ViT-B/16	66.40	71.61	74.83	75.52	-	-	-	-
EVA-02 ViT-B/16	69.63	72.64	74.64	75.52	-	-	-	-
DINO ViT-B/16	73.61	79.41	81.17	81.90	-	-	-	-
SAM ViT-B/16	65.05	70.40	71.58	71.95	-	-	-	-



Other Projects – Chair



Other Projects

- Privacy-preserving datasets by capturing distributions with CVAEs
- Exploring and evaluating deep hashing methods for similarity search and privacy preservation
- Human activity recognition via deep learning based on active exoskeleton data
- Component identification for geometry measurements in the vehicle development process using machine learning
- Deep learning-based evaluation of handwriting legibility using a sensor enhanced ballpoint pen
- Developing a task-agnostic data augmentation method via foreground extraction, object relocation, and generative inpainting

Interested?



Interested?



Website



Final Theses